Authentication over Arbitrarily Varying Channels with Causal Adversaries

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Abstract

We study authentication over a discrete memoryless arbitrarily varying channel (AVC) in which the adversary selects the channel state causally based on past channel outputs. We prove that the authentication capacity is positive exactly when the channel is not *distribution-overwritable* under stochastic encoding, and not *I-overwritable* under deterministic encoding. In the stochastic case, whenever the authentication capacity is positive, it coincides with the no-adversary Shannon capacity. The positivity results in both settings are obtained by constructing positive-rate codes with controlled overlap between codewords, together with a martingale concentration argument that handles adaptive adversarial strategies. For stochastic encoding, the full-capacity achievability further combines a capacity-achieving channel code with an authentication tag. These characterizations separate the causal stochastic-code and causal deterministic-code settings from each other and from the oblivious-adversary setting studied by Kosut and Kliewer.

I. Introduction

In an authentication problem, the receiver either decodes the transmitted message with confidence that it was not tampered with, or raises an alarm. We consider authentication over an arbitrarily varying channel (AVC) $W_{Y|X,S}$ in which the state S is chosen by an adversary observing the channel output causally (Fig. 1).

For reliable communication over an AVC, it is shown in [1] that the deterministic capacity is positive iff $W_{Y|X,S}$ is not *symmetrizable*. An analogous characterization for authentication is due to Kosut and Kliewer [2], [3]: the authentication capacity against an oblivious adversary is positive iff $W_{Y|X,S}$ is not *overwritable*, where $W_{Y|X,S}$ is overwritable if there is a conditional distribution $\mathsf{Q}_{S|X'}$ such that, for every $x, x' \in \mathscr{X}$,

$$\sum_{s\in\mathscr{S}} \mathsf{Q}_{S|X'}(s\,|\,x')\,W(y\,|\,x,s) = W(y\,|\,x',s_0) \quad \forall y\in\mathscr{Y}. \tag{1}$$

We refer to this as *obliviously overwritable*, to distinguish it from the related conditions below. Sufficient conditions for positive capacity were later established for myopic adversaries with non-causal noisy observations: $W_{Y|X,S}$ not I-overwritable [4], and $W_{Y|X,S}$ not distribution-overwritable [5].

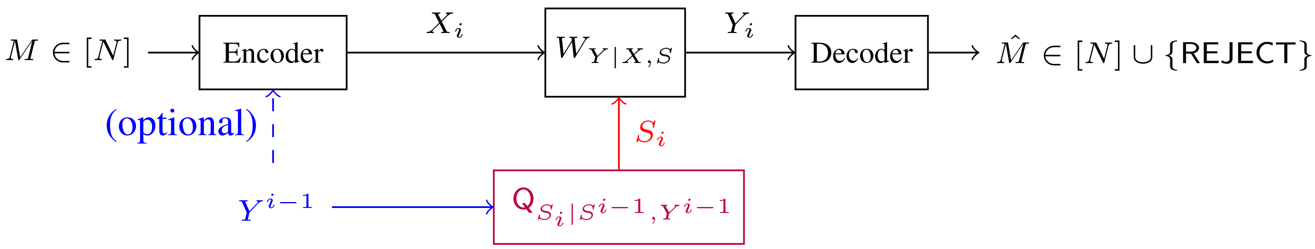

Fig. 1: Channel model. The adversary observes the channel output Y^{i-1} causally and selects state S_i. The dashed feedback link to the encoder is optional.

Our work

We consider an *output-feedback* adversary that, at each time t, chooses the state S_t as a function of (S^{t-1}, Y^{t-1}). This is the causal counterpart of the omniscient adversary and refines the oblivious adversary of [2]. The corresponding causal-adversary AVC reliability problem is treated in [6]–[9].

Other related work

Authentication has been studied beyond the present adversary model. It originates in cryptography [10]–[12] and has been considered on noisy channels with shared keys, in keyless schemes, and in Byzantine multiple-access settings [13]–[19]. Within the AVC framework, variants of the Kosut–Kliewer characterization include AWGN authentication [20], structured codes [21], and additional myopic and multiple-access models [22], [23]. Two-phase constructions in this line are related to identification codes [24]–[27], which also underpin our achievability through the doubly-exponential message growth of the authentication tag. The wait-and-overwrite converse is connected to the line of adversarial hypothesis testing [28]–[31].

Main results and organization

We consider the authentication capacity with both stochastic and deterministic codes against a causal adversary with output-feedback. Our results are stated in terms of two channel conditions strictly weaker than (1), called *distribution-* and *I-overwritability* (formalized in Section III). We show that:

(*i*) the stochastic authentication capacity is positive iff $W_{Y|X,S}$ is not distribution-overwritable;

(*ii*) the deterministic authentication capacity is positive iff $W_{Y|X,S}$ is not I-overwritable;

(*iii*) whenever the stochastic authentication capacity is positive, it equals the no-adversary Shannon capacity C_{s_0}.

The deterministic-code capacity, when positive, remains open, as it does in the omniscient setting [4]. Table I places these results alongside the oblivious, myopic, and omniscient dichotomies. A key contribution of our work is a wait-and-overwrite attack based on the posterior guessing probability of the message; this enables the converse for both the stochastic and deterministic codes. Our achievability borrows ideas from [5], modified for feedback adversaries through a martingale concentration argument.

Sections II–III state the model and formal results, Section IV gives the converse, and Sections V–VI sketch the achievability. Detailed proofs and examples are in the appendices.

TABLE I: Authentication-capacity dichotomies.

Adversary's view	$C_{\mathrm{auth,stoch}}=0$ iff	$C_{\mathrm{auth,det}}=0$ iff	Ref.
None (oblivious)	oblivious-OW	oblivious-OW	[2], [3]
Z^n (non-causal myopic)	myopic distribution-OW (multi-letter)	open	[4], [22], [5], [32]
X^n (omniscient)	I-OW	I-OW	[4]
(Y^{t-1}) (causal f.b.)	*distribution-OW*	*I-OW*	***Thms. 5, 7***

II. Preliminaries

A. Notation

For a positive integer N, $[N] \triangleq \{1,\dots,N\}$. Script letters $\mathscr{X},\mathscr{Y},\mathscr{S},\mathscr{M},\dots$ denote finite alphabets, capital letters the corresponding random variables, lowercase letters their realizations, and $y^t=(y_1,\dots,y_t)$. The set of probability distributions on $\mathscr{A}$ is $\mathcal{P}(\mathscr{A})$, $\mathbf{1}_a\in\mathcal{P}(\mathscr{A})$ is the point mass on a, and $\mathbb{V}(\cdot,\cdot)$ is the variational distance.

B. Channel and adversary

$W_{Y|X,S}$ is a discrete memoryless channel with finite alphabets $\mathscr{X},\mathscr{S},\mathscr{Y}$ and a fixed *no-adversary state* $s_0\in\mathscr{S}$; $W_{Y|X,S=s_0}$ is the legitimate channel and C_{s_0} its Shannon capacity. At time t the encoder emits $X_t\in\mathscr{X}$, the adversary picks $S_t\in\mathscr{S}$, and $Y_t\sim W(\cdot\mid X_t,S_t)$. The receiver outputs $\hat{M}\in[N]\cup\{\mathsf{REJECT}\}$, where REJECT flags the adversary's presence. Throughout, $M\in[N]$ is uniformly distributed.

C. Codes

An (N,n) *code* is a sequential encoder $F=\left\{\mathsf{P}_{X_t|M,X^{t-1},Y^{t-1}}\right\}_{t=1}^n$ paired with a decoder $\phi\colon\mathscr{Y}^n\to[N]\cup\{\mathsf{REJECT}\}$. The code is *deterministic* if each conditional law is a Dirac mass, and *stochastic* otherwise. The encoder is in the *feedback* (resp. *no-feedback*) class according to whether X_t may depend on Y^{t-1}.

D. Adversarial strategies and error metric

An *adversarial strategy* is a sequence $\mathsf{Q}=\left\{\mathsf{Q}_{S_t|S^{t-1},Y^{t-1}}\right\}_t$; we write Q_0 for the trivial strategy $S_t\equiv s_0$. The *authentication error* of a code (F,ϕ) is

$$\varepsilon(F,\phi)\triangleq\Pr_{F,\phi,\mathsf{Q}_0}(\hat{M}\neq M)+\sup_{\mathsf{Q}}\Pr_{F,\phi,\mathsf{Q}}(\hat{M}\notin\{M,\mathsf{REJECT}\}).$$

A rate R is *achievable* if there exist $(2^{nR},n)$ codes with $\varepsilon\to0$, and $C^{\bullet}_{\mathrm{auth},\circ}$ denotes the resulting capacity for $\bullet\in\{\mathrm{fb},\text{no-fb}\}$, $\circ\in\{\det,\mathrm{stoch}\}$. We write $\epsilon_n\triangleq\varepsilon(F,\phi)$; since ϵ_n upper-bounds the no-adversary error, any bound stated in terms of the latter also holds with ϵ_n.

E. Posterior-guessing probability

Our converse uses the *guessing probability*

$$\mathsf{G}_{A|B}(b)\triangleq\sum_{a\in\mathscr{A}}\Big(\Pr_{A|B}(a\mid b)\Big)^2,$$

specifically $\mathsf{G}_{M|Y^t,S^t}(y^t,s_0^t)$ under the no-adversary law.

III. Main results

A. Overwritability conditions

Our analysis uses two conditions strictly weaker than oblivious overwritability (1). Each says the adversary can pick states so the attacked channel matches the no-adversary channel $W(\cdot\,|\,x', s_0)$, at the indicated level of granularity.

Definition 1 (Distribution-overwritable)**.** $W_{Y|X,S}$ is *distribution-overwritable* if for every $\mathsf{P}\in\mathcal{P}(\mathscr{X})$ and every $x'\in\mathscr{X}$ there is $\mathsf{Q}\in\mathcal{P}(\mathscr{S})$ such that

$$\sum_{x,s}\mathsf{P}(x)\mathsf{Q}(s)\,W(y\,|\,x,s) = W(y\,|\,x', s_0) \quad \forall y\in\mathscr{Y}.$$

Definition 1 coincides with the single-letter form of the *myopic distribution-overwritability* of [5] when the adversary's side channel is trivial; the general myopic case has additional subtleties [5], [32].

Definition 2 (I-overwritable [4])**.** $W_{Y|X,S}$ is *I-overwritable* if for every $x,x'\in\mathscr{X}$ there is $\mathsf{Q}\in\mathcal{P}(\mathscr{S})$ such that

$$\sum_{s\in\mathscr{S}}\mathsf{Q}(s)\,W(y\,|\,x,s) = W(y\,|\,x', s_0) \quad \forall y\in\mathscr{Y}.$$

Remark 3 (Strict hierarchy). Obliviously overwritable $\Rightarrow$ distribution-overwritable $\Rightarrow$ I-overwritable. As noted in [5], [32] both implications are strict: a bit-flip channel is I-overwritable but not distribution-overwritable (Appendix F), and the AZBSC(δ,α) of [32] is distribution-overwritable but not obliviously overwritable (Appendix G).

Remark 4 (Non-overwritability witness). Since the alphabets are finite, the involved sets of distributions are compact. Hence, when $W_{Y|X,S}$ is not distribution-overwritable there exist $\mathsf{P}_X\in\mathcal{P}(\mathscr{X})$, $x'\in\mathscr{X}$, and $\delta>0$ with

$$\min_{\mathsf{Q}\in\mathcal{P}(\mathscr{S})} \mathbb{V}\big(\textstyle\sum_{x,s}\mathsf{P}_X(x)\mathsf{Q}(s)W(\cdot\,|\,x,s), W(\cdot\,|\,x',s_0)\big)\geq\delta; \tag{2}$$

similarly, when $W_{Y|X,S}$ is not I-overwritable there exist $x_0, x_0'\in\mathscr{X}$ and $\delta>0$ with

$$\min_{\mathsf{Q}\in\mathcal{P}(\mathscr{S})} \mathbb{V}\big(\textstyle\sum_{s}\mathsf{Q}(s)W(\cdot\,|\,x_0,s), W(\cdot\,|\,x_0',s_0)\big)\geq\delta. \tag{3}$$

We call $(\mathsf{P}_X, x', \delta)$ the *distribution-overwritability witness* and (x_0, x_0', δ) the *I-overwritability witness*; both are used by the achievability constructions in Sections V–VI.

B. Authentication capacity and positivity conditions

Theorem 5 (Positivity for stochastic codes)**.** *The following are equivalent:*

(i) $C^{\text{no-fb}}_{\text{auth,stoch}} > 0$*;*

(ii) $C^{\text{fb}}_{\text{auth,stoch}} > 0$*;*

(iii) $W_{Y|X,S}$ *is not distribution-overwritable.*

Theorem 6 (Stochastic capacity characterization)**.** *Whenever* $C^{\bullet}_{\text{auth,stoch}}>0$*,*

$$C^{\bullet}_{\text{auth,stoch}} = C_{s_0} \quad \textit{for } \bullet\in\{\text{fb}, \text{no-fb}\}\,.$$

Theorem 7 (Positivity for deterministic codes)**.** *The following are equivalent:*

(i) $C^{\text{no-fb}}_{\text{auth,det}} > 0$*;*

(ii) $C^{\text{fb}}_{\text{auth,det}} > 0$*;*

(iii) $W_{Y|X,S}$ *is not I-overwritable.*

Algorithm 1: Wait-and-overwrite attack.

Input: Encoder $\left\{\mathsf{P}_{X_t|M,X^{i-1},Y^{i-1}}\right\}$, threshold γ, observations $y_1,\ldots,y_n$.

```
/* Wait phase                                                              */
```

$t \leftarrow 1$;

while $\mathsf{G}_{M|Y^{t-1},S^{t-1}}(y^{t-1},s_0^{t-1}) \le \gamma$ *and* $t \le n$ **do**

  $s_t \leftarrow s_0$; $t \leftarrow t+1$;

end while

$T_\gamma \leftarrow t-1$; sample $\hat{m},\tilde{m} \overset{\text{i.i.d.}}{\sim} \mathsf{P}^{\mathrm{na}}_{M|Y^t,S^t}(\cdot\,|\,y^{T_\gamma},s_0^{T_\gamma})$;

```
/* Overwrite phase                                                         */
```

for $\tau = T_\gamma+1,\ldots,n$ **do**

  Sample $s_\tau \sim \mathsf{Q}^{(\tau)}$ so that the marginal at step τ matches a no-adv transmission of $\tilde{m}$;

end for

Remark 8. For deterministic codes our results give only positivity. The achievable rate is not known to equal C_{s_0}: after the wait phase, the adversary can guess the intended codeword and act effectively *omniscient*, but the omniscient-adversary capacity for deterministic codes is itself open. Determining $C^{\bullet}_{\mathrm{auth,det}}$ is left as an open problem. This mirrors a similar open question in the AVC reliability literature, where the capacity of deterministic codes under omniscient adversaries is not known in general [6], [9], [33].

IV. The wait-and-overwrite attack

A single attack proves both converse halves. The adversary plays $S_t = s_0$ during a *wait* phase while tracking the no-adversary posterior over M; once the posterior concentrates, it samples a guess and decoy from the posterior and enters an *overwrite* phase that forces the output to mimic a no-adversary transmission of the decoy. Distribution-OW gives the converse of Theorem 5, I-OW that of Theorem 7.

A. The attack

Fix a code (F,ϕ) and a threshold $\gamma \in (0, 1/(32|\mathscr{Y}|^2))$. During the wait phase the adversary sets $S_t = s_0$ until the no-adversary posterior $\mathsf{G}_{M|Y^{t-1},S^{t-1}}$ exceeds γ; let T_γ be the first such index. The adversary then samples a guess $\hat{M}$ and a decoy $\tilde{M}$ i.i.d. from the posterior at T_γ and, in the overwrite phase, plays a per-step strategy $\mathsf{Q}^{(\tau)}$ so the marginal law of Y_τ matches a no-adversary transmission of $\tilde{M}$ under the encoder's use of $\hat{M}$. The pseudocode is given in Algorithm 1.

B. Stopping-time lemma

The wait phase rests on the following stopping-time control of the posterior. Let M be uniform on $[N]$ and Y^n be the channel outputs under $S^n = s_0^n$.

Lemma 9 (Stopping time)**.** *Fix* $\gamma \in (0, 1/(32|\mathscr{Y}|^2))$ *and let*

$$T_\gamma \triangleq \inf\left\{t \in [n] : \mathsf{G}_{M|Y^t,S^t}(Y^t, s_0^t) > \gamma\right\},$$

with the convention $\inf\emptyset = \infty$*. For any* (N,n) *stochastic feedback code with no-adversary error probability* p_e*,*

$$\Pr(T_\gamma > n \mid S^n = s_0^n) \le \frac{p_e}{1-\sqrt{\gamma}}, \tag{4}$$

$$\Pr\!\left(\mathsf{G}_{M|Y^{T_\gamma},S^{T_\gamma}} > \tfrac{1}{2} \;\middle|\; T_\gamma \le n,\, S^{T_\gamma} = s_0^{T_\gamma}\right) \le \sqrt{2}\,|\mathscr{Y}|\sqrt{\gamma} < \tfrac{1}{4}. \tag{5}$$

Equation (4) says reliability of the code ($p_e \to 0$) forces $T_\gamma \le n$ with high probability. Equation (5) says that on the stopping event the posterior at T_γ stays below $1/2$ with probability $\ge 3/4$, which is the non-degeneracy that yields both $\hat{M} \ne M$ and $\tilde{M} \ne M$ with constant probability.

Proof sketch: For (4), $g(y^n) \ge (1-p^*(y^n))^2$ where p^* is the conditional MAP error, so $g(Y^n) \le \gamma$ forces $p^*(Y^n) \ge 1-\sqrt{\gamma}$, and Markov on $\mathbb{E}[p^*(Y^n)] \le p_e$ gives the bound. For (5), on $\{T_\gamma = t\}$ minimality gives $g(Y^{t-1}) \le \gamma$, and a sub-multiplicative bound on $\sqrt{\mathsf{G}}$ (Appendix A) combined with Markov yields the displayed inequality. Full proof in Appendix B.

C. Proof of converses

The wait-and-overwrite attack (Algorithm 1) runs the same way in both the stochastic and deterministic cases. The adversary samples $\hat{m}, \tilde{m}$ i.i.d. from the posterior at T_γ, commits to the hypothesis $M = \hat{m}$, and at each step $\tau > T_\gamma$ chooses the per-step state distribution $\mathsf{Q}^{(\tau)}$ as a function of the realized history $(\hat{m}, \tilde{m}, y^{\tau-1}, s^{\tau-1})$ so that the conditional law of Y_τ under the attack matches the no-adversary conditional law of Y_τ under $M = \tilde{m}$ given the same past:

– under distribution-overwritability (stochastic codes), let $\tilde{\mathsf{P}}^{(\tau)}$ be the encoder's conditional input distribution at step τ given $\left(M = \hat{m}, y^{\tau-1}, s^{\tau-1}\right)$ (computed by the path likelihood over $x^{\tau-1}$), and let $\mathsf{P}^{(\tau)}$ be the encoder's no-adversary conditional input distribution at step τ given $\left(M = \tilde{m}, y^{\tau-1}, s_0^{\tau-1}\right)$. By distribution-overwritability[1] there is a $\mathsf{Q}^{(\tau)}$ with

$$\sum_{x,s} \tilde{\mathsf{P}}^{(\tau)}(x) \mathsf{Q}^{(\tau)}(s) W_{Y|X,S}(\cdot \mid x, s) = \sum_{x'} \mathsf{P}^{(\tau)}(x') W_{Y|X,S}(\cdot \mid x', s_0).$$

The precise per-step construction of $\mathsf{Q}^{(\tau)}$ is detailed in Appendix C.

– under I-overwritability (deterministic codes), the codeword symbols $c_{\hat{m},\tau}, c_{\tilde{m},\tau}$ are determined by the respective messages and the realized $y^{\tau-1}$ (under encoder feedback), and $\mathsf{Q}^{(\tau)}$ satisfies the per-symbol identity

$$\sum_s \mathsf{Q}^{(\tau)}(s) W(\cdot \,|\, c_{\hat{m},\tau}, s) = W(\cdot \,|\, c_{\tilde{m},\tau}, s_0);$$

The success event for the attack is therefore $\{\hat{m} = M, \ \tilde{m} \ne M\}$.

By Lemma 9, under $\mathcal{E} \triangleq \left\{T_\gamma \le n, \ \mathsf{G}_{M|Y^{T_\gamma}} \le 1/2\right\}$, we have $\Pr(\mathcal{E}) \ge 3/4 - \epsilon_n/(1-\sqrt{\gamma})$. Since $\hat{m}, \tilde{m}, M$ are conditionally i.i.d. from the posterior given Y^{T_γ}, the success event has conditional probability $\mathsf{G}(1-\mathsf{G})$, which is at least $\gamma(1-\gamma)$ under $\mathcal{E}$. Conditioned on the success event, the channel output is a no-adversary transcript of $\tilde{m} \ne M$, so reliability of the code makes the decoder output $\tilde{m}$ with probability $\ge 1-\epsilon_n$ – an authentication error. Multiplying gives the bound:

$$\Pr_{\text{adv}}\left(\hat{M} \notin \{M, \mathsf{REJECT}\}\right) \ge \gamma(1-\gamma)(1-\epsilon_n)\left[\frac{3}{4} - \frac{\epsilon_n}{1-\sqrt{\gamma}}\right]. \tag{6}$$

Proposition 10 (Converse halves of Theorems 5 and 7)**.** *Fix $\gamma \in (0, 1/(32|\mathscr{Y}|^2))$. For any blocklength-$n$ feedback code (F, ϕ) with auth-error ϵ_n, the wait-and-overwrite attack satisfies* (6) *provided either*

(i) (F, ϕ) is a stochastic code and $W_{Y|X,S}$ is distribution-overwritable, or

(ii) (F, ϕ) is a deterministic code and $W_{Y|X,S}$ is I-overwritable.

In particular, $C^{\text{fb}}_{\text{auth,stoch}} = 0$ in case (i) and $C^{\text{fb}}_{\text{auth,det}} = 0$ in case (ii).

[1]For every $\mathsf{P}, \mathsf{P}' \in \mathcal{P}(\mathscr{X})$, averaging the per-$x'$ witnesses of Definition 1 against $\mathsf{P}'(x')$ yields a Q with $\mathbb{E}_{X,S} W_{Y|X,S}(\cdot \,|\, X, S) = \mathbb{E}_{X' \sim \mathsf{P}'} W_{Y|X,S}(\cdot \,|\, X', s_0)$.

Remark 11. The two overwritability conditions affect *which* channels admit the attack, not the success probability. The bottleneck in both cases is the event $\{\hat{m}=M\}$, of conditional probability $\geq\gamma$ on $\mathcal{E}$, giving the $\gamma(1-\gamma)$ factor in (6).

Remark 12 (Encoder feedback does not change the positivity of the capacity). Proposition 10 uses only the encoder's input *distribution* under feedback, not the absence of encoder feedback. The same attack applies verbatim with or without encoder feedback, so both cases of the proposition cover C^{fb} and $C^{\text{no-fb}}$.

V. Positivity proofs for Theorems 5 and 7

We sketch the achievability of Theorem 5; Theorem 7 follows along similar lines. Fix $(\lambda_1,\lambda_2)\in(0,1)^2$. Our construction adapts the overlapping-sets scheme of [5] to the output-feedback adversary via a flip-set martingale concentration. We build a code with rate $\frac{1}{n}\log N_n^{(\text{auth})}\geq R^{(\text{auth})}>0$, no-adversary error $\leq\lambda_1$, and missed-detection error $\leq\lambda_2$ uniformly over feedback adversaries Q, for all n large enough:

<u>*Step 1: Codebook:*</u> Fix $\alpha\in(0,1)$ and a slack $\beta\in(0,1)$. By [5, Lemma 3], build a family $\mathfrak{B}=\{\mathscr{B}_m\subset[n]\}_{m\in[N]}$ of subsets of $[n]$ with $N\geq 2^{R^{(\text{auth})}n}$, in which each set has size in $\alpha(1\pm\beta)n$, distinct sets have intersection at most $\alpha^2(1+\beta)n$, and an *overlap property* that we use in Step 3.

<u>*Step 2: Encoder/Decoder:*</u> Pick a non-overwritability witness (P_X,x',δ) from (2). The encoder transmits $X_t=x'$ on $\mathscr{B}_m$ and $X_t\sim\mathsf{P}_X$ i.i.d. off $\mathscr{B}_m$. Given Y^n, for any $\mathscr{B}\subseteq[n]$ let $\widehat{P}_{Y^n}(\cdot\,|\,\mathscr{B})$ denote the empirical type of Y^n restricted to positions $t\in\mathscr{B}$, i.e. $\widehat{P}_{Y^n}(y\,|\,\mathscr{B})=\frac{1}{|\mathscr{B}|}\sum_{t\in\mathscr{B}}\mathbf{1}\{Y_t=y\}$. The decoder computes

$$T_{\hat{m}}(Y^n)\triangleq\mathbb{V}\big(\widehat{P}_{Y^n}(\cdot\,|\,\mathscr{B}_{\hat{m}}),W(\cdot\,|\,x',s_0)\big)$$

for each $\hat{m}\in[N]$ and outputs the unique $\hat{m}$ with $T_{\hat{m}}(Y^n)\leq\delta/4$, or REJECT when none or several candidates pass.

<u>*Step 3: Decoding-error bound:*</u> The decoder errs only when some wrong $\hat{m}\neq M$ also passes the test. We show that for each fixed $m\neq\hat{m}$ and any feedback adversary Q,

$$T_{\hat{m}}(Y^n)\geq\delta/2\quad\text{with probability } 1-e^{-\Omega(n)},\tag{7}$$

and a union bound over $N\leq 2^{R^{(\text{auth})}n}$ candidates closes the argument once $R^{(\text{auth})}$ is chosen small enough. The overlap property of Step 1 reduces (7) to a TV bound on the flip set $J\triangleq\mathscr{B}_{\hat{m}}\setminus\mathscr{B}_m$:

$$\mathbb{V}\big(\widehat{P}_{Y^n}(\cdot\,|\,J),W(\cdot\,|\,x',s_0)\big)\geq\delta/2.\tag{8}$$

A per-letter analysis cannot proceed directly: S_t depends on Y^{t-1}, so the Y_t's on the flip set are neither independent nor identically distributed, and the strategy $\{\mathsf{Q}_{S_t|\mathscr{F}_{t-1}}\}$ ranges over an uncountable path-dependent class. Non-overwritability (2) sidesteps this: for *every* single-letter $Q\in\mathcal{P}(\mathscr{S})$, the averaged channel $\sum_{x,s}\mathsf{P}_X(x)Q(s)W(\cdot\,|\,x,s)$ is δ-far in TV from $W(\cdot\,|\,x',s_0)$. On the flip set the encoder draws fresh i.i.d. P_X, so with $\mathscr{F}_{t-1}\triangleq\sigma(M,X^t,S^t,Y^t)$ and $Q=\mathsf{Q}_{S_t|\mathscr{F}_{t-1}}$, the conditional law π_t of Y_t given $\mathscr{F}_{t-1}$ satisfies $\mathbb{V}(\pi_t,W(\cdot\,|\,x',s_0))\geq\delta$ on every history. The centred sequence $Z_t\triangleq\mathbf{1}\{Y_t\in A\}-\pi_t(A)$ is then a bounded-difference $\{\mathscr{F}_t\}$-martingale with zero conditional mean regardless of Q, and Azuma–Hoeffding gives (8) with probability $1-e^{-\Omega(n)}$ and an exponent depending only on $(\delta,\alpha,\beta,|\mathscr{Y}|)$. The supremum over Q moves outside the probability.

A. Achievability for deterministic codes (Theorem 7)

We prove the achievability direction of Theorem 7 as the specialization $\mathsf{P}_X=\delta_{x_0}$ of the stochastic auth-code construction of Section V; if $W_{Y|X,S}$ is not I-overwritable then $C^{\bullet}_{\text{auth,det}}>0$. Concretely, the witness (P_X,x',δ) becomes $(\delta_{x_0},x_0',\delta)$, with δ now the *I-overwritability* gap of (3). The key differences from the stochastic case are in three places:

- *Encoder.* With $\mathsf{P}_X = \delta_{x_0}$, the encoder is fully deterministic: $X_t = x_0'$ for $t \in \mathscr{B}_m$ and $X_t = x_0$ for $t \notin \mathscr{B}_m$. The codeword is binary, $c_m \in \{x_0, x_0'\}^n$.
- *Witness gap.* The dist-OW gap in step (1) of the auth-tag proof reduces, since P_X is a Dirac mass, to the I-OW gap (3), namely
$$\inf_{\mathsf{Q}} \mathbb{V}\big(\textstyle\sum_s \mathsf{Q}(s) W(\cdot \,|\, x_0, s), W(\cdot \,|\, x_0', s_0)\big) \geq \delta.$$
- *Step (2) martingale.* On the flip set the encoder transmits x_0 deterministically; the conditional Bernoulli mean is $p_t = \sum_s \mathsf{Q}_{S_t|\mathscr{F}_{t-1}}(s \mid \mathscr{F}_{t-1}) W(A \,|\, x_0, s)$, δ-far from $W(A \,|\, x_0', s_0)$ uniformly over Q. Azuma–Hoeffding on the same bounded-difference martingale gives the same flip-set concentration.

The codebook (overlapping-sets lemma) and the overlap property (step (3)) are unchanged – they are statements about set geometry and do not depend on the input distribution. Combining gives $C^{\bullet}_{\text{auth,det}} \geq R > 0$ for both $\bullet \in \{\text{fb}, \text{no-fb}\}$. The full deterministic proof is in Appendix E.

Remark 13. Let $C^* \triangleq \max_{\mathsf{P}_X \in \mathcal{P}(\{x_0, x_0'\})} I(X; Y \,|\, S = s_0)$ be the Shannon capacity of the no-adversary sub-channel $W(y \,|\, x, s_0)$ under inputs restricted to $\{x_0, x_0'\}$. The proof yields a positive rate strictly below C^*. The gap to C^* (and a fortiori to C_{s_0}) is an open problem – see Remark 8.

VI. Capacity for stochastic codes (Theorem 6)

The auth-code of Section V only guarantees an exponential number of messages, short of C_{s_0}. We reach the no-adversary capacity in two steps: (i) build an *authentication tag* with doubly-exponentially many candidates by composing the auth-code with an Ahlswede–Dueck identification code for the noiseless binary channel (Section VI-A); (ii) concatenate the tag with a capacity-achieving channel code on $W_{Y|X,S=s_0}$, with the channel code carrying M in n_1 symbols and the tag carrying a verification statistic in $n_2 = O(\log n)$ symbols, giving composite rate $\frac{nR}{n_1+n_2} \to R$ (Section VI-B).

A. Authentication tags

An (N, n)-*authentication tag* [5, Def. 9] consists of a stochastic encoder $\mathsf{P}_{X^n|M}$ and a family of deterministic verifiers $\{D_{\hat{m}} : \mathscr{Y}^n \to \{\mathsf{ACCEPT}, \mathsf{REJECT}\}\}_{\hat{m}}$ indexed by a candidate message $\hat{m}$ already available to the receiver, with low false-alarm and missed-detection error (formal definition in Appendix D). The tag's *rate* is the doubly-exponential growth $\frac{1}{n} \log \log N_n$.

Lemma 14 (Existence of authentication tags)**.** *If $W_{Y|X,S}$ is not distribution-overwritable, then there exists $R > 0$ such that, for every $(\lambda_1, \lambda_2) \in (0,1)^2$ and all n sufficiently large, there is an (N_n, n)-tag with errors (λ_1, λ_2) uniformly over all output-feedback adversary strategies, and $\liminf_{n\to\infty} \frac{1}{n} \log \log N_n \geq R$.*

The proof has two parts. First, we take an authentication code of Section V, whose missed-detection bound is uniform over output-feedback adversaries. Next, we compose this auth-code with an Ahlswede–Dueck identification code [25] for the noiseless binary channel: each auth-code codeword is reinterpreted as the output of a length-$\tilde{n}$ identification code with $\lfloor 2^{2^{\tilde{n}\tilde{R}}} \rfloor$ candidate messages, and the verifier on a candidate $\hat{m}$ first runs the auth-code's decoder and then the identification verifier on the decoded codeword. The identification step is not exposed to the adversary, so no further feedback handling is needed at the lift. The result is a tag with $\frac{1}{n} \log \log N_n \geq \tilde{R}\hat{R} = R$. The full construction is in Appendix D-B.

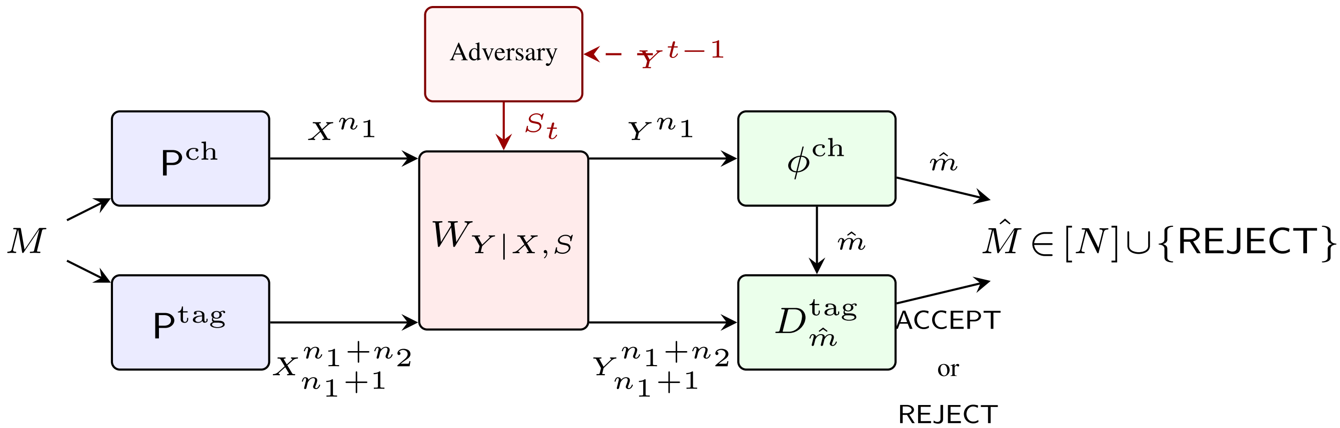

Fig. 2: Code construction for Theorem 6.

B. Concatenation with a channel code

The achievability scheme concatenates a capacity-achieving channel code with the tag of Lemma 14 (Fig. 2).

Fix $R < C_{s_0}$. Choose blocklength n so that:

(*i*) there is an $(\lceil 2^{nR}\rceil, n_1)$ channel code $(\mathsf{P}^{\mathrm{ch}}_{X^{n_1}|M}, \phi^{\mathrm{ch}})$ over the no-adversary channel $W_{Y|X,S=s_0}$ with average error $\epsilon_n = o(1)$;
(*ii*) there is an $(\lceil 2^{nR}\rceil, n_2)$ authentication tag $(\mathsf{P}^{\mathrm{tag}}_{X^{n_2}|M}, \{D^{\mathrm{tag}}_{\hat{m}}\})$ of error pair (ϵ_n, ϵ_n) for $W_{Y|X,S}$, guaranteed by Lemma 14 as long as $n_2 = \Omega(\log\log\lceil 2^{nR}\rceil) = O(\log n)$.

The composite encoder uses

$$\mathsf{P}_{X^{n_1+n_2}|M}(x^{n_1+n_2}|m) = \mathsf{P}^{\mathrm{ch}}_{X^{n_1}|M}(x^{n_1}|m)\mathsf{P}^{\mathrm{tag}}_{X^{n_2}|M}(x^{n_1+n_2}_{n_1+1}|m).$$

The decoder first runs ϕ^{ch} on Y^{n_1} to obtain a candidate $\hat{m} = \phi^{\mathrm{ch}}(Y^{n_1})$, then verifies it via the tag:

$$\phi(Y^{n_1+n_2}) = \begin{cases} \hat{m} & \text{if } D^{\mathrm{tag}}_{\hat{m}}(Y^{n_1+n_2}_{n_1+1}) = \mathsf{ACCEPT}, \\ \mathsf{REJECT} & \text{otherwise.} \end{cases} \tag{9}$$

Error analysis: Under no adversary, the channel-code stage outputs $\hat{m} = M$ with probability $\geq 1-\epsilon_n$, and the verifier accepts the correct tag with probability $\geq 1-\epsilon_n$. So $\Pr_{\mathrm{na}}(\phi \neq M) \leq 2\epsilon_n$.

Under any adversary strategy, the channel-code stage outputs *some* candidate $\hat{m} \in [N]$. If $\hat{m} = M$, the tag verifier accepts and authentication succeeds. If $D^{\mathrm{tag}}_{\hat{m}}(Y^{n_1+n_2}_{n_1+1}) = \mathsf{REJECT}$, the composite decoder outputs REJECT and authentication succeeds. Otherwise $\hat{m} \neq M$ *and* the verifier accepts – but this is exactly the missed-detection event of the tag, whose probability is at most ϵ_n for every adversary strategy. Combining,

$$\Pr_{\mathrm{adv}}(\phi(Y^{n_1+n_2}) \notin \{M, \mathsf{REJECT}\}) \leq \epsilon_n.$$

Hence the composite code achieves authentication error $\leq 3\epsilon_n = o(1)$ at rate $\frac{nR}{n+O(\log n)} \to R$ as $n \to \infty$.

Since $R < C_{s_0}$ was arbitrary, $C^{\mathrm{fb}}_{\mathrm{auth,stoch}} \geq C_{s_0}$, and combined with the trivial reverse inequality this proves Theorem 6.

Remark 15. During the wait phase the adversary may decode M from Y^{n_1} and become effectively M-omniscient in the tag phase. The tag's missed-detection bound nevertheless holds uniformly: the encoder's fresh i.i.d. P_X on the flip set $\mathscr{B}_{\hat{m}} \setminus \mathscr{B}_m$ hides X_t at each position, and the non-distribution-overwritability gap (2) controls π_t uniformly over $\mathsf{Q}_{S_t|\mathscr{F}_{t-1}}$. Details are in Appendix D-C.

VII. Conclusion

We characterized the authentication capacity of an AVC against a causal adversary with output feedback. For stochastic codes, positivity is equivalent to $W_{Y|X,S}$ not being distribution-overwritable, and whenever the capacity is positive it equals the no-adversary Shannon capacity C_{s_0}. For deterministic codes, positivity is equivalent to $W_{Y|X,S}$ not being I-overwritable; characterizing the deterministic-code capacity remains open, paralleling the analogous gap in the omniscient-adversary setting. The converses follow from a single wait-and-overwrite attack driven by the posterior guessing probability, and the achievability uses an overlapping-set code analyzed via a flip-set martingale that absorbs the adversary's adaptivity without a union bound over strategies. Reaching C_{s_0} requires composing this authentication code with an Ahlswede–Dueck identification tag, to obtain an authentication tag and, concatenating the resulting authentication tag with a channel code. Natural directions for future work include extending the framework to continuous alphabets, to feedback adversaries with delayed or noisy observations, and to multi-user settings.

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TABLE II: Proof roadmap. Building blocks for the converse and achievability are listed first; the main theorems combine the two sides.

Result	Proved in	Uses
Converse-side building blocks		
Lemma 16 (sub-multiplicativity)	App. A	—
Lemma 9 (stopping time)	App. B	Lemma 16
Claim 17 (per-step matching)	App. C	dist-OW (Def. 1) / I-OW (Def. 2)
Proposition 10 (converse halves)	App. C	Lemma 9, Claim 17
Achievability-side building blocks		
Auth-code construction (single-exp.)	Section V, App. D-A	non-distribution-OW gap (2), flip-set martingale
Lemma 14 (auth-tag, doubly-exp.)	App. D-B	auth-code construction, identification codes
Concatenation with channel code	App. D-C	Lemma 14, capacity-achieving channel code on $W_{Y\mid X,S=s_0}$
Main theorems (combining both sides)		
Theorem 5 (stochastic positivity)	Achiev.: Section V; Conv.: App. C, case (i)	auth-code construction, Proposition 10
Theorem 6 (stochastic capacity)	Achiev.: Section VI, App. D; Conv.: App. C	Lemma 14, Proposition 10
Theorem 7 (deterministic positivity)	Achiev.: Section V-A, App. E; Conv.: App. C, case (ii)	Const.-comp. codebook + I-OW, Proposition 10
Illustrative examples (Remark 3)		
Bit-flip channel	App. F	I-overwritable but not distribution-overwritable
AZBSC(δ, α)	App. G	distribution-overwritable but not obliviously overwritable

Appendix A
Sub-multiplicative growth of the guessing probability

Lemma 16 (Guessing-probability sub-multiplicativity)**.** *Under the no-adversary law* na*,*

$$\mathbb{E}\Big[\sqrt{\mathsf{G}_{M\mid Y^t,S^t}(Y^t,s_0^t)}\ \Big|\ Y^{t-1}=y^{t-1}\Big] \le |\mathscr{Y}|\sqrt{\mathsf{G}_{M\mid Y^{t-1},S^{t-1}}(y^{t-1},s_0^{t-1})}.$$

Proof. Write $g(y^t)\triangleq \mathsf{G}_{M\mid Y^t,S^t}(y^t,s_0^t)$ and $p(y_t\,|\,y^{t-1})\triangleq \mathsf{P}_{Y_t\mid Y^{t-1},S^t}(y_t\,|\,y^{t-1},s_0^t)$. For every (m,y_t),

$$\mathsf{P}_{M,Y_t\mid Y^{t-1}}(m,y_t\,|\,y^{t-1}) \le \mathsf{P}_{M\mid Y^{t-1}}(m\,|\,y^{t-1}).$$

Squaring and summing over m gives

$$\sum_m \mathsf{P}_{M,Y_t\mid Y^{t-1}}(m,y_t\,|\,y^{t-1})^2 \le g(y^{t-1}).$$

Therefore

$$\mathbb{E}[\sqrt{g(Y^t)}\,|\,y^{t-1}] = \sum_{y_t} p(y_t\,|\,y^{t-1})\sqrt{\textstyle\sum_m \mathsf{P}_{M\mid Y^t}(m\,|\,y^t)^2}$$

$$= \sum_{y_t} \sqrt{\textstyle\sum_m \big(p(y_t \,|\, y^{t-1}) \mathsf{P}_{M|Y^t}(m \,|\, y^t)\big)^2} = \sum_{y_t} \sqrt{\textstyle\sum_m \mathsf{P}_{M,Y_t|Y^{t-1}}(m, y_t \,|\, y^{t-1})^2} \le \sum_{y_t} \sqrt{g(y^{t-1})} = |\mathscr{Y}| \sqrt{g(y^{t-1})}.$$

□

APPENDIX B
PROOF OF LEMMA 9

Throughout this appendix all probabilities and expectations are taken under the no-adversary law na, i.e., the joint law of (M, X^n, Y^n) when the adversary plays $S^n = s_0^n$. Write $g(y^t) \triangleq \mathsf{G}_{M|Y^t,S^t}(y^t, s_0^t)$ and recall $T_\gamma = \inf\{t \in [n] : g(Y^t) > \gamma\}$.

Proof of (4). Let $p_e^{(n)} \triangleq \Pr_{\mathrm{na}}(\hat{M} \ne M)$, and write $p^*(y^n) \triangleq 1 - \max_m \mathsf{P}_{M|Y^n}(m \,|\, y^n)$ for the conditional MAP error given $Y^n = y^n$. Since the MAP rule is optimal, $\mathbb{E}[p^*(Y^n)] \le p_e^{(n)}$. The guessing probability dominates the squared MAP success probability:

$$g(y^n) \ge \big(\max_m \mathsf{P}_{M|Y^n}(m \,|\, y^n)\big)^2 = (1 - p^*(y^n))^2,$$

so $g(Y^n) \le \gamma$ forces $p^*(Y^n) \ge 1 - \sqrt{\gamma}$. By Markov,

$$\Pr(T_\gamma > n) \le \Pr(g(Y^n) \le \gamma) \le \frac{p_e^{(n)}}{1 - \sqrt{\gamma}} \xrightarrow[n \to \infty]{} 0.$$

Proof of (5). Under the event $\{T_\gamma = t\}$, the definition of T_γ as the *first* crossing time gives $g(Y^{t-1}) \le \gamma$. Combining with Lemma 16,

$$\mathbb{E}\big[\sqrt{g(Y^t)} \;\big|\; T_\gamma = t\big] \le |\mathscr{Y}| \sqrt{\gamma}.$$

Markov's inequality with $\gamma < 1/(32|\mathscr{Y}|^2)$ yields

$$\begin{aligned} \Pr\big(g(Y^t) > \tfrac{1}{2} \;\big|\; T_\gamma = t\big) &\le \sqrt{2} \cdot \mathbb{E}\big[\sqrt{g(Y^t)} \;\big|\; T_\gamma = t\big] \\ &\le \sqrt{2}\, |\mathscr{Y}| \sqrt{\gamma} < \tfrac{1}{4}. \end{aligned}$$

Averaging over $t \in [n]$ weighted by $\Pr(T_\gamma = t)$ gives $\Pr(g(Y^{T_\gamma}) > 1/2 \,|\, T_\gamma \le n) < 1/4$, i.e., (5). □

APPENDIX C
DETAILED PROOF OF THE CONVERSE

We give a single detailed proof of the converse Proposition 10, covering both the stochastic/distribution-overwritable and the deterministic/I-overwritable cases. We first establish the per-step matching identity (Claim 17); then write the attack joint distribution explicitly; decompose the resulting distribution into a "matched" part ($M = \hat{M}$) and a "mismatched" part ($M \ne \hat{M}$); bound the rejection probability and the correct-decoding probability separately; and finally combine via the high-probability event $\mathcal{E}$.

Throughout, (F, ϕ) is a feedback code of blocklength n with auth-error ϵ_n, $\mathscr{D}_\perp \triangleq \phi^{-1}(\mathsf{REJECT})$, and the wait-and-overwrite attack of Section IV stops at T_γ and overwrites with $\tilde{M}$. We use $\tilde{\mathsf{P}}$ for the joint distribution under the attack and P_{na} for the no-adversary joint. The argument is written for the stochastic case; the deterministic case is the specialization in which $\tilde{\mathsf{P}}^{(\tau)} = \mathbf{1}_{c_{\hat{m},\tau}}$ is a point mass and dist-OW reduces to I-OW (cf. end of this appendix).

Per-step matching:

Claim 17. *Suppose $W_{Y|X,S}$ is distribution-overwritable. Fix $\tau \in [T_\gamma + 1 : n]$ and a history $h^{\tau-1} \triangleq (\hat{m}, \tilde{m}, y^{\tau-1}, s^{\tau-1})$. Let $\tilde{\mathsf{P}}^{(\tau)}$ be the encoder's marginal at step τ under the attack with $M = \hat{M} = \hat{m}$, and let $\mathsf{P}^{(\tau)}$ be the encoder's marginal at step τ under no-adversary with $M = \tilde{m}$. Then there exists $\mathsf{Q}^{(\tau)} \in \mathcal{P}(\mathscr{S})$ with*

$$\sum_{x_\tau, s_\tau} \tilde{\mathsf{P}}^{(\tau)}(x_\tau)\mathsf{Q}^{(\tau)}(s_\tau)W_{Y|X,S}(y_\tau \,|\, x_\tau, s_\tau) = \sum_{x'_\tau} \mathsf{P}^{(\tau)}(x'_\tau)W_{Y|X,S}(y_\tau \,|\, x'_\tau, s_0) \tag{10}$$

for every $y_\tau \in \mathscr{Y}$.

Proof. $\tilde{\mathsf{P}}^{(\tau)}$ is computable from the encoder's conditional law and the realized $(\hat{m}, y^{\tau-1}, s^{\tau-1})$ via the path likelihood

$$\mathsf{P}_{X_\tau|M,Y^{\tau-1},S^{\tau-1}}(x_\tau \,|\, m, h^{\tau-1}) = \frac{\Lambda_\tau(m, x_\tau, h^{\tau-1})}{\sum_{x'_\tau} \Lambda_\tau(m, x'_\tau, h^{\tau-1})},$$

where

$$\Lambda_\tau(m, x_\tau, h^{\tau-1}) = \sum_{x^{\tau-1}} \left(\prod_{i=1}^{\tau-1} \mathsf{P}_{X_i|M,X^{i-1},Y^{i-1}}(x_i \,|\, m, x^{i-1}, y^{i-1}) \times W_{Y|X,S}(y_i \,|\, x_i, s_i) \right) \times \mathsf{P}_{X_\tau|M,X^{\tau-1},Y^{\tau-1}}(x_\tau \,|\, m, x^{\tau-1}, y^{\tau-1}).$$

Averaging the per-x' witnesses of Definition 1 against $\mathsf{P}^{(\tau)}(x')$ with $\mathsf{P} = \tilde{\mathsf{P}}^{(\tau)}$ yields a $\mathsf{Q}^{(\tau)}$ satisfying (10). □

Telescoping: For $i \in [T_\gamma + 1 : n]$, define the hybrid trajectory

$$\Gamma_i(y^n_{T_\gamma+1} \,|\, \hat{m}, \tilde{m}, y^{T_\gamma}) \triangleq A_i \cdot B_i,$$

where

$$A_i \triangleq \sum_{s^i_{T_\gamma+1}} \prod_{\tau=T_\gamma+1}^{i} \tilde{\mathsf{P}}_{Y_\tau,S_\tau|M,\hat{M},\tilde{M},Y^{\tau-1},S^{\tau-1}}(y_\tau, s_\tau \,|\, \dots),$$

$$B_i \triangleq \prod_{\tau=i+1}^{n} \sum_{x_\tau} \mathsf{P}_{X_\tau|M,Y^{\tau-1},S^{\tau-1}}(x_\tau \,|\, \tilde{m}, y^{\tau-1}, s_0^{\tau-1}) \times W_{Y|X,S}(y_\tau \,|\, x_\tau, s_0).$$

A_i records the actual attack up through step i, and B_i the no-adversary distribution thereafter. Substituting (10) into the inner sum over s_i inside A_i collapses the $\tau = i$ factor of A_i onto the $\tau = i$ factor of B_{i-1}, so $\Gamma_i = \Gamma_{i-1}$. Iterating from $i = n$ down to $i = T_\gamma + 2$ yields the telescoping identity

$$\Gamma_n = \Gamma_{T_\gamma+1}. \tag{11}$$

Explicit attack joint distribution:

For $t \in [n]$ and $y^n \in \mathscr{G}_t = \{T_\gamma(y^n) = t\}$, the wait phase forces $S^t = s_0^t$ and the overwrite phase samples $\hat{M}, \tilde{M}$ i.i.d. from $\mathsf{P}^{\mathrm{na}}_{M|Y^t,S^t}(\cdot \,|\, y^t, s_0^t)$ and uses the strategies $\{\mathsf{Q}^{(\tau)}\}_{\tau > t}$. Hence

$$\begin{aligned}
&\tilde{\mathsf{P}}(M = m, \hat{M} = \hat{m}, \tilde{M} = \tilde{m}, Y^n = y^n, S^n = (s_0^t, s_{t+1}^n)) \\
&= \mathsf{P}_M(m)\, \mathsf{P}^{\mathrm{na}}_{Y^t|M,S^t}(y^t \,|\, m, s_0^t)\, \mathsf{P}^{\mathrm{na}}_{M|Y^t,S^t}(\hat{m} \,|\, y^t, s_0^t) \\
&\quad \cdot \mathsf{P}^{\mathrm{na}}_{M|Y^t,S^t}(\tilde{m} \,|\, y^t, s_0^t) \\
&\quad \cdot \tilde{\mathsf{P}}_{Y^n_{t+1},S^n_{t+1}|M,\hat{M},\tilde{M},Y^t,S^t}(y^n_{t+1}, s^n_{t+1} \,|\, m, \hat{m}, \tilde{m}, y^t, s_0^t).
\end{aligned} \tag{12}$$

Term-by-term decomposition:

Marginalize (12) over $(\hat{m}, m, s^n_{t+1})$ and split on $m = \hat{m}$ vs $m \neq \hat{m}$:

$$\tilde{\mathsf{P}}(Y^n = y^n) = \mathsf{P}^{\mathrm{na}}_{Y^t|S^t}(y^t \,|\, s_0^t)\left[A_{\mathrm{term}}(y^n) + B_{\mathrm{term}}(y^n)\right],$$

where

$$A_{\text{term}}(y^n) \triangleq \sum_{\hat m,\tilde m}\big(\mathsf{P}^{\text{na}}_{M|Y^t,S^t}(\hat m\,|\,y^t,s_0^t)\big)^2\cdot \mathsf{P}^{\text{na}}_{M|Y^t,S^t}(\tilde m\,|\,y^t,s_0^t)\,\Gamma_n,$$

$$B_{\text{term}}(y^n) \triangleq \sum_{\substack{m,\tilde m\\ \hat m\neq m}}\mathsf{P}^{\text{na}}_{M|Y^t,S^t}(m\,|\cdot)\,\mathsf{P}^{\text{na}}_{M|Y^t,S^t}(\hat m\,|\cdot)\cdot\mathsf{P}^{\text{na}}_{M|Y^t,S^t}(\tilde m\,|\cdot)\,\Gamma_n.$$

By the telescoping identity (11), $\Gamma_n=\Gamma_{T_\gamma+1}$, which is independent of $\hat m$ and equals the no-adversary likelihood of Y_{t+1}^n given $M=\tilde m$ and $Y^t=y^t$. Substituting into A_{term} and using $\sum_{\hat m}(\mathsf{P}^{\text{na}}_{M|Y^t,S^t}(\hat m\,|\cdot))^2=\mathsf{G}_{M|Y^t,S^t}(y^t,s_0^t)$,

$$A_{\text{term}}(y^n)=\mathsf{G}_{M|Y^t,S^t}(y^t,s_0^t)\times\mathsf{P}^{\text{na}}_{Y_{t+1}^n|Y^t,S^n}(y_{t+1}^n\,|\,y^t,s_0^n). \tag{13}$$

Multiplying by $\mathsf{P}^{\text{na}}_{Y^t|S^t}(y^t\,|\,s_0^t)$ and using $\mathsf{G}_{M|Y^t,S^t}\le 1$,

$$\mathsf{P}^{\text{na}}_{Y^t|S^t}(y^t\,|\,s_0^t)\,A_{\text{term}}(y^n)\le \Pr_{\text{na}}(Y^n=y^n). \tag{14}$$

For B_{term}, marginalize over $\tilde m$ first ($\sum_{\tilde m,y_{t+1}^n}\mathsf{P}^{\text{na}}_{M|Y^t,S^t}\,\Gamma_n=1$); then $\sum_{\hat m\neq m}\mathsf{P}^{\text{na}}_{M|Y^t,S^t}(\hat m\mid\cdot)=1-\mathsf{P}^{\text{na}}_{M|Y^t,S^t}(m\mid\cdot)$, and $\sum_m\mathsf{P}^{\text{na}}_{M|Y^t,S^t}(m\,|\cdot)(1-\mathsf{P}^{\text{na}}_{M|Y^t,S^t}(m\,|\cdot))=1-\mathsf{G}_{M|Y^t,S^t}(y^t,s_0^t)$. Hence

$$\sum_{y_{t+1}^n}B_{\text{term}}(y^n)=1-\mathsf{G}_{M|Y^t,S^t}(y^t,s_0^t)\le 1-\gamma\quad\text{for } y^t\in\mathscr{F}_t, \tag{15}$$

by the stopping condition $\mathsf{G}_{M|Y^t,S^t}>\gamma$ on $\mathscr{F}_t$.

Total bound on $\tilde{\mathsf{P}}(\mathscr{D}_\perp)$:

Combining (14) and (15) and summing over t,

$$\begin{aligned}\tilde{\mathsf{P}}(\mathscr{D}_\perp)&=\sum_{t=1}^n\sum_{y^n\in\mathscr{G}_t\cap\mathscr{D}_\perp}\tilde{\mathsf{P}}(Y^n=y^n)\\ &\le\Pr_{\text{na}}(\mathscr{D}_\perp)+(1-\gamma)\sum_t\Pr_{\text{na}}(T_\gamma=t)+\Pr_{\text{na}}(T_\gamma>n)\\ &\le\epsilon_n+(1-\gamma)+\tfrac{\epsilon_n}{1-\sqrt{\gamma}},\end{aligned}$$

where the last step uses Lemma 9(4) for the residual mass on $\{T_\gamma>n\}$ and $\Pr_{\text{na}}(\mathscr{D}_\perp)\le\epsilon_n$.

Total bound on $\tilde{\mathsf{P}}(\hat M=M)$:

Repeat the decomposition above but now sum y^n over the event $\{\phi(y^n)=m\}$ rather than $\mathscr{D}_\perp$. After using the same telescoping identity (i.e., (13) absorbed into a no-adversary likelihood for message $\tilde m$), and splitting on $m=\tilde m$ vs $m\neq\tilde m$,

$$\begin{aligned}\tilde{\mathsf{P}}(\hat M=M)&\le\sum_{m,t}\mathsf{P}_M(m)\,\mathsf{P}^{\text{na}}_{M|Y^t,S^t}(m\,|\,y^t,s_0^t)+\epsilon_n\\ &\le\mathbb{E}_{\text{na}}\big[\mathsf{G}_{M|Y^t,S^t}(Y^{T_\gamma},s_0^{T_\gamma})\big]+\epsilon_n.\end{aligned}$$

By Lemma 9(5), $\mathsf{G}_{M|Y^t,S^t}\le 1/2$ with conditional probability $\ge 3/4$ on $\{T_\gamma\le n\}$ and $\mathsf{G}_{M|Y^t,S^t}\le 1$ otherwise; together with (4),

$$\mathbb{E}_{\text{na}}[\mathsf{G}_{M|Y^t,S^t}(Y^{T_\gamma},s_0^{T_\gamma})]\le\tfrac{5}{8}+\tfrac{\epsilon_n}{1-\sqrt{\gamma}}.$$

Combining:

The events $\Big\{\hat M=\mathsf{REJECT}\Big\}$ and $\Big\{\hat M=M\Big\}$ are disjoint, so

$$\tilde{\mathsf{P}}(\hat M\notin\{M,\mathsf{REJECT}\})=1-\tilde{\mathsf{P}}(\hat M=\mathsf{REJECT})-\tilde{\mathsf{P}}(\hat M=M)\ge\gamma-\tfrac{5}{8}-2\epsilon_n-\tfrac{2\epsilon_n}{1-\sqrt{\gamma}}. \tag{16}$$

Bound (16) is loose because γ is small. The tighter bound used in Proposition 10 comes from conditioning on the high-probability event $\mathcal{E}=\{T_\gamma\le n, \mathsf{G}_{M|Y^t,S^t}\le 1/2\}$. Since $M,\hat{M},\tilde{M}$ are conditionally i.i.d. from the posterior given Y^{T_γ}, and since $x\mapsto x(1-x)$ is increasing on $[0,1/2]$,

$$\tilde{\mathsf{P}}\big(\hat{M}=M,\ \tilde{M}\neq M\,\big|\,\mathcal{E}\big)=\mathsf{G}_{M|Y^t,S^t}(1-\mathsf{G}_{M|Y^t,S^t})\ge\gamma(1-\gamma).$$

On $\mathcal{E}\cap\Big\{\hat{M}=M,\ \tilde{M}\neq M\Big\}$ the matching of Claim 17 (or, in the deterministic case, the I-OW per-symbol match) makes the channel output a no-adversary transcript of $\tilde{M}$, and reliability gives $\Pr_{\mathrm{na}}(\phi(Y^n)=\tilde{M}\,|\,M=\tilde{M})\ge 1-\epsilon_n$, so the decoder produces an authentication error with conditional probability $\ge\gamma(1-\gamma)(1-\epsilon_n)$ on $\mathcal{E}$. By Lemma 9, $\Pr_{\mathrm{adv}}(\mathcal{E})\ge 3/4-\epsilon_n/(1-\sqrt{\gamma})$. Multiplying,

$$\tilde{\mathsf{P}}(\hat{M}\notin\{M,\mathsf{REJECT}\})\ge\gamma(1-\gamma)(1-\epsilon_n)\left[\tfrac{3}{4}-\tfrac{\epsilon_n}{1-\sqrt{\gamma}}\right],\tag{17}$$

which is (6). $\square$

Specialization to the deterministic case:

For deterministic codes, the encoder's marginal $\tilde{\mathsf{P}}^{(\tau)}$ at step τ is a point mass at $c_{\hat{m},\tau}$. The matching condition (10) then reduces to the per-symbol identity $\sum_s \mathsf{Q}^{(\tau)}(s)W(\cdot\,|\,c_{\hat{m},\tau},s)=W(\cdot\,|\,c_{\tilde{m},\tau},s_0)$, which is exactly I-overwritability for the pair $(c_{\hat{m},\tau},c_{\tilde{m},\tau})$. The rest of the proof is identical, and the same bound (17) holds.

Encoder feedback does not change the proof:

The matching at step τ uses $\tilde{\mathsf{P}}^{(\tau)}=\tilde{\mathsf{P}}_{X_\tau|M=\hat{m},\hat{m},\tilde{m},Y^{\tau-1},S^{\tau-1}}$, which the adversary can compute from $(\hat{m},y^{\tau-1},s^{\tau-1})$. A feedback encoder that, having access to $Y^{\tau-1}$ as well, makes X_τ depend on the past does not change this — the adversary, having access to $Y^{\tau-1}$ as well, can compute the encoder's distribution exactly. Hence the same proof covers both C^{fb} and $C^{\text{no-fb}}$.

Remark 18 (Sharpness). The constant $\frac{3\gamma(1-\gamma)}{4}$ is an artifact of Markov's inequality at threshold $1/2$ in Lemma 9 and the loose bound $\mathsf{G}(1-\mathsf{G})\ge\gamma(1-\gamma)$ on $\mathcal{E}$. Iterating the stopping rule, using a tighter Doob argument on $\sqrt{\mathsf{G}(Y^t,s_0^t)}$, or accounting for the actual joint $\mathsf{G}-\sum_m P(m)^3$ can improve the constant; the qualitative conclusion is unchanged.

Appendix D

Detailed proofs for stochastic achievability

Definition 19 (((N,n)-Authentication tag [5, Def. 9])**.** An (N,n)-*authentication tag* for $W_{Y|X,S}$ consists of a stochastic encoder $\mathsf{P}_{X^n|M}\colon[N]\to\mathcal{P}(\mathscr{X}^n)$ together with a family of deterministic verifiers $\{D_{\hat{m}}\colon\mathscr{Y}^n\to\{\mathsf{ACCEPT},\mathsf{REJECT}\}\}_{\hat{m}\in[N]}$, indexed by a candidate $\hat{m}\in[N]$ already available to the receiver. The tag achieves error pair (λ_1,λ_2) if

$$\max_{m\in[N]}\Pr\big(D_m(Y^n)=\mathsf{REJECT}\,\big|\,M=m,\ S^n=s_0^n\big)\le\lambda_1,$$

$$\max_{\substack{m\in[N]\\ \hat{m}\neq m}}\sup_{\mathsf{Q}}\Pr\big(D_{\hat{m}}(Y^n)=\mathsf{ACCEPT}\,\big|\,M=m\big)\le\lambda_2,$$

where Q ranges over feedback adversary strategies. The *tag capacity* C_{tag} is the supremum over R such that for every $(\lambda_1,\lambda_2)\in(0,1)^2$ there is a sequence of (N_n,n)-tags achieving (λ_1,λ_2) with $\liminf_n\frac{1}{n}\log\log N_n\ge R$.

A. Auth-code construction: detailed proof of the achievability of Theorem 5

We use the same code construction as in [5, Sec. V], based on the overlapping-sets lemma [5, Lemma 3]. The construction is a single template that handles both the deterministic and the stochastic cases through one parameter setting; we state it in stochastic form (general witness distribution P_X). The deterministic case (Section V-A) is the specialization $\mathsf{P}_X = \delta_{x_0}$ called out at the end of this proof.

Non-overwritability witness: Since $W_{Y|X,S}$ is not distribution-overwritable, by Remark 4 (Eq. (2)) there exist $\mathsf{P}_X \in \mathcal{P}(\mathscr{X})$, $x' \in \mathscr{X}$, and $\delta > 0$ with

$$\kappa(\mathsf{P}_X, x') \triangleq \min_{\mathsf{Q}\in\mathcal{P}(\mathscr{S})} \mathbb{V}\big(\textstyle\sum_{x,s} \mathsf{P}_X(x)\mathsf{Q}(s)W(\cdot\,|\,x,s),\, W(\cdot\,|\,x',s_0)\big) \geq \delta. \tag{18}$$

The pair (P_X, x') is the *non-overwritability witness*: P_X is the encoder's witness input distribution off the verification set, and x' is the target symbol against which the verifier compares.

Codebook (overlapping-sets lemma): By [5, Lemma 3] (which builds on the general Gilbert bound for constant-composition codes, [34, Problem 10.1]), for any $\alpha, \beta \in (0,1)$ and any rate $R < \min\{\beta^2\alpha^2/6,\, \beta^2\alpha(1-\alpha)/4\}$ there exists a family $\mathfrak{B} = \{\mathscr{B}_m \subseteq [n] : m \in [N]\}$ with $N \geq 2^{Rn}$ satisfying:

(i) $\alpha(1-\beta)n \leq |\mathscr{B}_m| \leq \alpha(1+\beta)n$ for all m;

(ii) $|\mathscr{B}_m \cap \mathscr{B}_{\hat{m}}| < \alpha^2(1+\beta)n$ for all $m \neq \hat{m}$;

(iii) $|\mathscr{B}_{\hat{m}} \setminus \mathscr{B}_m| > \alpha(1-\alpha)(1+\beta)n$ for all $m \neq \hat{m}$;

(iv) *(Overlap property.)* For every $y^n \in \mathscr{Y}^n$ and every $m, \hat{m}$,

$$\mathbb{V}\big(\widehat{P}_{y^n}(\cdot\,|\,\mathscr{B}_{\hat{m}}),\, W(\cdot\,|\,x',s_0)\big) \geq \tfrac{(1-\alpha)(1-\beta)}{1+\beta}\, \mathbb{V}\big(\widehat{P}_{y^n}(\cdot\,|\,\mathscr{B}_{\hat{m}} \setminus \mathscr{B}_m),\, W(\cdot\,|\,x',s_0)\big) \quad - \tfrac{2\alpha(1+\beta)}{1-\beta}, \tag{19}$$

where $\widehat{P}_{y^n}(\cdot\,|\,\mathscr{B})$ is the empirical type of y^n on positions $t \in \mathscr{B}$. Property (iv) is what handles the match-vs-flip cancellation in the analysis below: the TV at the full verification set is bounded below by the TV at the flip set $\mathscr{B}_{\hat{m}} \setminus \mathscr{B}_m$ alone, modulo a small additive correction tunable through α, β.

Encoder: Conditioned on $M = m$:

$$X_t = \begin{cases} x' & \text{if } t \in \mathscr{B}_m, \\ \sim \mathsf{P}_X \text{ i.i.d.} & \text{if } t \notin \mathscr{B}_m, \end{cases} \tag{20}$$

i.e., on the message-specific subset $\mathscr{B}_m$ the encoder transmits x' deterministically, and off that subset it draws i.i.d. from the witness distribution P_X.

Verifier: Given candidate $\hat{m}$ and received Y^n, accept iff

$$\mathbb{V}\big(\widehat{P}_{Y^n}(\cdot\,|\,\mathscr{B}_{\hat{m}}),\, W(\cdot\,|\,x',s_0)\big) \leq \delta/4. \tag{21}$$

Choice of α, β: Pick $\alpha, \beta \in (0,1)$ small enough that

$$\tfrac{(1-\alpha)(1-\beta)}{1+\beta} \cdot \tfrac{\delta}{2} - \tfrac{2\alpha(1+\beta)}{1-\beta} > \tfrac{\delta}{4}. \tag{22}$$

This is achievable since the LHS tends to $\delta/2 > \delta/4$ as $\alpha, \beta \to 0$.

No-adversary analysis (false alarm): Fix $m = \hat{m}$ and $S^n = s_0^n$. For $t \in \mathscr{B}_m$ the encoder transmits x' deterministically and $Y_t \sim W(\cdot\,|\,x',s_0)$ i.i.d. By Hoeffding plus a $|\mathscr{Y}|$-fold union bound,

$$\Pr\big(\mathbb{V}(\widehat{P}_{Y^n}(\cdot\,|\,\mathscr{B}_m), W(\cdot\,|\,x',s_0)) > \delta/4\big) \leq 2|\mathscr{Y}|e^{-c_1 n} \tag{23}$$

for some $c_1 > 0$ depending only on $\alpha, \delta, |\mathscr{Y}|$. Union-bounded over $N \leq 2^{nR}$ messages, the false-alarm error is $o(1)$ provided $R \ln 2 < c_1$.

Adversarial analysis (missed detection): Fix $m \neq \hat{m}$ and a feedback adversary $\mathsf{Q} = \{\mathsf{Q}_{S_t|S^{t-1},Y^{t-1}}\}$. By the overlap property (19) and the choice of α, β in (22), acceptance of $\hat{m}$ (i.e., (21)) implies

$$\mathbb{V}\big(\widehat{P}_{Y^n}(\cdot\,|\,\mathscr{B}_{\hat{m}} \setminus \mathscr{B}_m),\, W(\cdot\,|\,x',s_0)\big) < \delta/2. \tag{24}$$

We now bound the flip-set TV explicitly. Set $J_{\text{flip}} \triangleq \mathscr{B}_{\hat{m}} \setminus \mathscr{B}_m$ and let $N \triangleq |J_{\text{flip}}| > \alpha(1-\alpha)(1+\beta)n$ by (iii). Order the flip positions as $t_1 < t_2 < \cdots < t_N$.

The natural filtration adapted to the sequential adversary is

$$\mathscr{F}_t \triangleq \sigma\big(M, \hat{M}, X^t, S^t, Y^t\big).$$

On the flip set, $t \notin \mathscr{B}_m$, so the encoder transmits $X_t \sim \mathsf{P}_X$ i.i.d., conditionally independent of $\mathscr{F}_{t-1}$. The adversary then picks S_t from the kernel $\mathsf{Q}_{S_t|\mathscr{F}_{t-1}}(\cdot \mid \mathscr{F}_{t-1})$, and the channel output is $Y_t \sim W(\cdot \mid X_t, S_t)$. For any test set $A \subseteq \mathscr{Y}$ and any $t \in J_{\text{flip}}$,

$$\Pr\big(Y_t \in A \,|\, \mathscr{F}_{t-1}\big) = \sum_{x,s} \mathsf{P}_X(x)\, \mathsf{Q}_{S_t|\mathscr{F}_{t-1}}(s \,|\, \mathscr{F}_{t-1})\, W(A \,|\, x, s) =: p_t^{(A)}.$$

The kernel $\mathsf{Q}_{S_t|\mathscr{F}_{t-1}}$ encodes the adversary's sequential strategy. Note that $p_t^{(A)}$ is $\mathscr{F}_{t-1}$-measurable. By the non-overwritability gap (18),

$$\big|p_t^{(A)} - W(A \,|\, x', s_0)\big| \geq \delta \quad \text{a.s., uniformly over } \mathsf{Q}. \tag{25}$$

Note that (25) holds uniformly over Q, not on average: the gap is realized at each $\mathscr{F}_{t-1}$-history. This lets a single Azuma bound below cover all feedback adversaries simultaneously, sparing us a supremum over an uncountable strategy class inside the probability. We define the following bounded-difference martingale. Let

$$Z_t^{(A)} \triangleq \mathbf{1}\{Y_t \in A\} - p_t^{(A)}, \qquad t \in J_{\text{flip}},$$

and the partial sum $M_T^{(A)} \triangleq \sum_{k=1}^{T} Z_{t_k}^{(A)}$. Since $\mathbb{E}[Z_t^{(A)} \,|\, \mathscr{F}_{t-1}] = 0$ a.s. and $|Z_t^{(A)}| \leq 1$, $\left\{M_T^{(A)}\right\}$ is an $\{\mathscr{F}_{t_T}\}$-adapted martingale with bounded differences. Azuma–Hoeffding gives, for any $\xi > 0$,

$$\Pr\big(|M_N^{(A)}| > N\xi\big) \leq 2e^{-N\xi^2/2}. \tag{26}$$

Next, let $\widehat{P}_{Y^n}(A \,|\, J_{\text{flip}}) = \frac{1}{N}\sum_{k=1}^{N} \mathbf{1}\{Y_{t_k} \in A\}$ and $\bar{p}^{(A)} \triangleq \frac{1}{N}\sum_{k=1}^{N} p_{t_k}^{(A)}$, (26) reads

$$\Pr\big(|\widehat{P}_{Y^n}(A \,|\, J_{\text{flip}}) - \bar{p}^{(A)}| > \xi\big) \leq 2e^{-N\xi^2/2}. \tag{27}$$

Choose $A = A^*$ to be the TV-attaining set in (18); then (25) averaged over $t \in J_{\text{flip}}$ gives $|\bar{p}^{(A^*)} - W(A^* \mid x', s_0)| \geq \delta$ a.s. Picking $\xi = \delta/2$, the reverse triangle inequality in (27) yields

$$\Pr\big(|\widehat{P}_{Y^n}(A^* \,|\, J_{\text{flip}}) - W(A^* \,|\, x', s_0)| < \delta/2\big) \leq 2e^{-N\delta^2/8}.$$

Since $\mathbb{V}(P, Q) = \sup_A |P(A) - Q(A)|$, taking $A = A^*$ on the LHS,

$$\Pr\big(\mathbb{V}(\widehat{P}_{Y^n}(\cdot \,|\, J_{\text{flip}}), W(\cdot \,|\, x', s_0)) < \delta/2\big) \leq 2e^{-N\delta^2/8} \leq 2e^{-c_2 n}, \tag{28}$$

where $c_2 \triangleq \alpha(1-\alpha)(1+\beta)\delta^2/8 > 0$ uses $N > \alpha(1-\alpha)(1+\beta)n$ from (iii). Combining (24) and (28), missed detection has probability $\leq 2e^{-c_2 n}$ for any single $\hat{m} \neq m$. Union-bounded over $N \leq 2^{nR}$ candidates, missed detection is $o(1)$ provided $R \ln 2 < c_2$.

Conclusion: Let c_1 and c_2 satisfy (23) and (28) respectively and let α, β satisfy (22). The above analysis shows that as long as R satisfies $R < \min\left\{c_1/\ln 2,\, c_2/\ln 2,\, \beta^2\alpha^2/6,\, \beta^2\alpha(1-\alpha)/4\right\}$, both error events are $o(1)$ uniformly over feedback adversaries. $\square$

B. Proof of Lemma 14: lift via identification codes

We lift the auth-code of Appendix D-A to the doubly-exponential rate claimed in Lemma 14, mirroring the construction of [5]: compose the auth-code with an Ahlswede–Dueck identification code [25] for the noiseless binary channel.

Setup: Let W_{id} denote the noiseless binary channel ($\mathscr{X}=\mathscr{Y}=\{0,1\}$, output equals input). Fix $R<R^{(\mathrm{auth})}$ where $R^{(\mathrm{auth})}>0$ is the auth-code rate guaranteed by Appendix D-A. Pick $\hat{R}\in(R,R^{(\mathrm{auth})})$ and let $\tilde{R}\triangleq R/\hat{R}\in(0,1)$. Fix target errors $\lambda_1,\lambda_2\in(0,1/2)$.

Inner identification code: By [25, Theorem 1], for every $\tilde{n}$ large enough there is an $(N^{(\mathrm{id})},\tilde{n})$-identification code $\left(\widetilde{\mathsf{P}}_{B^{\tilde{n}}|M},\left\{\tilde{\phi}_m\right\}_{m\in[N^{(\mathrm{id})}]}\right)$ for W_{id} with misidentification errors $(\lambda_1/2,\lambda_2/2)$ and

$$N^{(\mathrm{id})}\geq\left\lfloor 2^{2^{\tilde{n}\tilde{R}}}\right\rfloor. \tag{29}$$

Here $M\in[N^{(\mathrm{id})}]$ is the source message, $\widetilde{\mathsf{P}}_{B^{\tilde{n}}|M}$ is the (stochastic) identification encoder, and $\tilde{\phi}_m:\{0,1\}^{\tilde{n}}\to\{\mathsf{ACCEPT},\mathsf{REJECT}\}$ is the candidate-m identification verifier. Its codeword set has size at most $N^{(\mathrm{id})}_{\mathrm{out}}\triangleq 2^{\tilde{n}}$.

Outer auth-code: By Appendix D-A, for every n large enough there is an $(N^{(\mathrm{auth})},n)$-*authentication code* $\left(\widehat{\mathsf{P}}_{X^n|B^{\tilde{n}}},\widehat{\Phi}\right)$ for $W_{Y|X,S}$ with no-adversary error $\leq\min\{\lambda_1/2,\lambda_2/2\}$, missed-detection error $\leq\min\{\lambda_1/2,\lambda_2/2\}$ uniformly over feedback adversaries, and

$$N^{(\mathrm{auth})}\geq\left\lfloor 2^{n\hat{R}}\right\rfloor.$$

The authentication code's message space is identified with the identification codeword space $\{0,1\}^{\tilde{n}}$ by choosing n so that $N^{(\mathrm{auth})}\geq N^{(\mathrm{id})}_{\mathrm{out}}=2^{\tilde{n}}$ (achievable since $\hat{R}>0$); the auth-code verifier $\widehat{\Phi}(y^n)\in\{0,1\}^{\tilde{n}}\cup\{\mathsf{REJECT}\}$ either outputs a decoded identification codeword or rejects.

Composite tag: For each candidate $\hat{m}\in[N^{(\mathrm{id})}]$, define an $(N^{(\mathrm{id})},n)$-tag for $W_{Y|X,S}$ with encoder

$$\mathsf{P}_{X^n|M}(x^n\,|\,m)=\sum_{b^{\tilde{n}}\in\{0,1\}^{\tilde{n}}}\widehat{\mathsf{P}}_{X^n|B^{\tilde{n}}}(x^n\,|\,b^{\tilde{n}})\,\widetilde{\mathsf{P}}_{B^{\tilde{n}}|M}(b^{\tilde{n}}\,|\,m),$$

and verifiers

$$D_{\hat{m}}(y^n)=\begin{cases}\mathsf{REJECT} & \text{if } \widehat{\Phi}(y^n)=\mathsf{REJECT},\\ \mathsf{REJECT} & \text{if } \widehat{\Phi}(y^n)=\hat{b}^{\tilde{n}}\in\{0,1\}^{\tilde{n}}\\ & \qquad\text{and } \tilde{\phi}_{\hat{m}}(\hat{b}^{\tilde{n}})=\mathsf{REJECT},\\ \mathsf{ACCEPT} & \text{otherwise.}\end{cases}$$

That is, the receiver first runs the auth-code verifier $\widehat{\Phi}$ on Y^n to decode either an identification codeword $\hat{b}^{\tilde{n}}$ or REJECT; on a successful auth-decode it then runs the identification verifier $\tilde{\phi}_{\hat{m}}$ for the candidate $\hat{m}$ on $\hat{b}^{\tilde{n}}$.

Rate: By (29) and the requirement $N^{(\mathrm{auth})}\geq 2^{\tilde{n}}$ (i.e., $\tilde{n}\leq(1/\hat{R})\log N^{(\mathrm{auth})}\leq n$),

$$\frac{1}{n}\log\log N^{(\mathrm{id})}\geq\frac{\tilde{n}\,\tilde{R}}{n}\geq\frac{\tilde{n}\,\tilde{R}}{(1/\hat{R})\log N^{(\mathrm{auth})}}\geq\tilde{R}\,\hat{R}=R.$$

Thus $\frac{1}{n}\log\log N_n\geq R$, matching the rate claim of Lemma 14.

Error analysis: For the no-adversary side, $S^n=s_0^n$ implies $\widehat{\Phi}(Y^n)=B^{\tilde{n}}$ except with probability $\leq\lambda_1/2$ (auth-code false alarm), and on that event the identification verifier $\tilde{\phi}_M(B^{\tilde{n}})$ accepts except with probability $\leq\lambda_1/2$ (identification false alarm). By union bound, the composite false alarm is $\leq\lambda_1$.

For the missed-detection side, fix $M=m$, $\hat{m}\neq m$, and any feedback adversary Q. The composite verifier accepts $\hat{m}$ only if both (i) $\widehat{\Phi}(Y^n)=\hat{b}^{\tilde{n}}\in\{0,1\}^{\tilde{n}}$ (auth-code does not reject), and (ii) $\tilde{\phi}_{\hat{m}}(\hat{b}^{\tilde{n}})=\mathsf{ACCEPT}$ (identification verifier accepts). On (i), the auth-code construction of Appendix D-A guarantees that $\hat{b}^{\tilde{n}}$ equals the encoded $B^{\tilde{n}}$ except on an event of Q-uniform probability $\leq \lambda_2/2$. On the complementary event $\hat{b}^{\tilde{n}} = B^{\tilde{n}}$, the identification verifier $\tilde{\phi}_{\hat{m}}(B^{\tilde{n}})$ accepts with probability $\leq \lambda_2/2$ (identification missed detection). Union-bounding the two events,

$$\sup_{\mathsf{Q}} \Pr\big(D_{\hat{m}}(Y^n) = \mathsf{ACCEPT} \bigm| M = m\big) \leq \lambda_2 \quad \forall \hat{m} \neq m.$$

C. Concatenation with a channel code

We now provide full analysis for the composite encoder of Section VI, addressing the subtlety that under the composite, the adversary in the tag phase may have *additionally* observed the wait-phase outputs Y^{n_1}.

Setup: Let $(F^{\mathrm{ch}}, \phi^{\mathrm{ch}})$ be a length-n_1 channel code over $W_{Y|X,S=s_0}$ at rate $R < C_{s_0}$ with no-adversary error $\epsilon_n = o(1)$. Let $(\mathsf{P}^{\mathrm{tag}}, \{D_{\hat{m}}^{\mathrm{tag}}\})$ be an (N_n, n_2)-tag from Lemma 14 with errors (ϵ_n, ϵ_n), $n_2 = O(\log n)$. The composite encoder is the product distribution (9); the composite decoder is $\phi(Y^{n_1+n_2}) = \hat{m}$ if $D_{\hat{m}}^{\mathrm{tag}}(Y_{n_1+1}^{n_1+n_2}) = \mathsf{ACCEPT}$ and REJECT otherwise, where $\hat{m} = \phi^{\mathrm{ch}}(Y^{n_1})$.

No-adversary error: Under $S^{n_1+n_2} = s_0^{n_1+n_2}$, the codeword stage outputs the correct $\hat{m} = M$ with probability $\geq 1 - \epsilon_n$ and, on that event, the tag stage accepts with conditional probability $\geq 1 - \epsilon_n$ (since the tag's false alarm is $\leq \epsilon_n$). By a union bound, $\Pr_{\mathrm{na}}(\phi(Y^{n_1+n_2}) \neq M) \leq 2\epsilon_n$.

Missed-detection error under feedback adversary: Fix any feedback adversary $\mathsf{Q} = \{\mathsf{Q}_{S_t|S^{t-1},Y^{t-1}}\}_{t=1}^{n_1+n_2}$. The composite missed-detection event is

$$\mathcal{M} \triangleq \left\{\phi(Y^{n_1+n_2}) \notin \{M, \mathsf{REJECT}\}\right\},$$

i.e. $\hat{m} = \phi^{\mathrm{ch}}(Y^{n_1}) \neq M$ *and* $D_{\hat{m}}^{\mathrm{tag}}(Y_{n_1+1}^{n_1+n_2}) = \mathsf{ACCEPT}$. We bound $\Pr_{\mathrm{adv}}(\mathcal{M})$ via two steps.

Step 1 (decompose by $\hat{m}$). Conditioning on (M, Y^{n_1}), the candidate $\hat{m} = \phi^{\mathrm{ch}}(Y^{n_1})$ is determined; on the event $\hat{m} \neq M$, the tag-phase outputs $Y_{n_1+1}^{n_1+n_2}$ form a length-n_2 tag-channel transcript with encoder $\mathsf{P}_{X^{n_2}|M}^{\mathrm{tag}}$ and adversary states $S_{n_1+1}^{n_1+n_2}$ drawn according to $\mathsf{Q}_{S_t|S^{t-1},Y^{t-1}}$ *restricted to the tag phase but with Y^{t-1} inheriting the wait-phase outputs Y^{n_1} as prefix.*

Step 2 (induced adversary in tag phase). For each fixed wait-phase realization (M, Y^{n_1}, S^{n_1}), define an *induced* tag-phase adversary

$$\widetilde{\mathsf{Q}}_{S_t|S_{n_1+1}^{t-1}, Y_{n_1+1}^{t-1}} \triangleq \mathsf{Q}_{S_t|S^{t-1},Y^{t-1}}\big(\cdot \,|\, (S^{n_1}, S_{n_1+1}^{t-1}), (Y^{n_1}, Y_{n_1+1}^{t-1})\big).$$

This $\widetilde{\mathsf{Q}}$ is a feedback tag-phase adversary in the sense of Definition 19 – its t-th conditional uses only history through time $t-1$ within the tag phase, with the wait-phase realization frozen as a parameter. Lemma 14 states that for *every* such tag-phase adversary, $\Pr(D_{\hat{m}}^{\mathrm{tag}} = \mathsf{ACCEPT} \mid M = m, \widetilde{\mathsf{Q}}) \leq \epsilon_n$ for all $\hat{m} \neq m$. Marginalizing over (M, Y^{n_1}, S^{n_1}) under the composite attack,

$$\begin{aligned}
&\Pr_{\mathrm{adv}}(\mathcal{M}) \\
&\quad = \mathbb{E}_{\mathrm{adv}}\big[\mathbf{1}_{\hat{m} \neq M} \, \Pr(D_{\hat{m}}^{\mathrm{tag}} = \mathsf{ACCEPT} \,|\, M, \hat{m}, Y^{n_1}, S^{n_1})\big] \\
&\quad \leq \mathbb{E}_{\mathrm{adv}}[\mathbf{1}_{\hat{m} \neq M} \, \epsilon_n] \leq \epsilon_n.
\end{aligned}$$

Total: The auth-error of the composite is the sum of the no-adversary error and the worst-case missed-detection:

$$\varepsilon(\phi) \leq \Pr_{\mathrm{na}}(\phi \neq M) + \sup_{\mathsf{Q}} \Pr_{\mathrm{adv}}(\mathcal{M}) \leq 3\epsilon_n = o(1).$$

The composite rate is $\frac{nR}{n_1+n_2} = \frac{nR}{n_1 + O(\log n)} \to R$, so $C_{\mathrm{auth,stoch}}^{\mathrm{fb}} \geq R$ for every $R < C_{s_0}$, finishing the achievability halves of Theorems 5 and 6. $\square$

APPENDIX E

PROOF OF DETERMINISTIC ACHIEVABILITY (THEOREM 7)

The deterministic achievability follows similar to the auth-code construction proved in Appendix D-A with P_X replaced by δ_{x_0}A. We list the only places where the proof differs and verify that the rest carries over verbatim.

Witness: Use the I-overwritability witness (x_0, x_0', δ) from (3). The distribution-overwritability gap (18) reduces to

$$\inf_{\mathsf{Q}\in\mathcal{P}(\mathscr{S})} \mathbb{V}\big(\textstyle\sum_s \mathsf{Q}(s)W(\cdot\,|\,x_0,s),\, W(\cdot\,|\,x_0',s_0)\big) \geq \delta,$$

which is exactly (3).

Encoder: Substituting $\mathsf{P}_X = \delta_{x_0}$ in (20) makes the encoder fully deterministic: $X_t = x_0'$ for $t \in \mathscr{B}_m$, $X_t = x_0$ otherwise. The codeword $c_m \in \{x_0, x_0'\}^n$ is a deterministic function of m.

Step 1 of Appendix D-A (conditional law on the flip set): On J_{flip} the encoder transmits $X_t = x_0$ deterministically, so the conditional output law given $\mathscr{F}_{t-1}$ specializes to

$$\pi_t(y) = \sum_{s\in\mathscr{S}} \mathsf{Q}_{S_t|\mathscr{F}_{t-1}}(s \mid \mathscr{F}_{t-1})\, W(y\,|\,x_0,s).$$

Its flip-set average $\bar{P} = \sum_s \bar{\mathsf{Q}}(s)W(\cdot\,|\,x_0,s)$ satisfies $\mathbb{V}(\bar{P}, W(\cdot\,|\,x_0',s_0)) \geq \delta$ a.s. by the I-overwritability gap.

Step 2 (martingale concentration): Unchanged. The bounded-difference martingale $Z_t = \mathbf{1}_{Y_t\in A} - \pi_t(A)$ on J_{flip} (with π_t specialized as above) and the Azuma–Hoeffding bound (28) apply verbatim, since the $|Z| \leq 1$ envelope and the conditional-mean centering do not depend on the input distribution.

Step 3 (overlap property): Unchanged. The overlap-property bound (19) is a statement about codebook geometry alone and is identical for both cases.

Conclusion: The combining argument of Appendix D-A carries over verbatim, giving $\Pr_{\text{adv}}(\hat{M} \notin \{m, \mathsf{REJECT}\}) \leq \lceil 2^{nR} \rceil \cdot 2|\mathscr{Y}|e^{-c_2\mu n} = o(1)$ for $R\ln 2 < c_2\mu$. Theorem 7's achievability follows. $\square$

APPENDIX F

BIT-FLIP CHANNEL: I-OVERWRITABLE BUT NOT DISTRIBUTION-OVERWRITABLE

The separation exhibited by this channel between I-overwritable and distribution-overwritable is established in [32]; we reprise the verification here for completeness.

Channel. Take $\mathscr{X} = \mathscr{Y} = \{0,1\}$, $\mathscr{S} = \{s_0, s_1\}$, with $W_{Y|X,S}(y\,|\,x,s_0) = \mathbf{1}\{y = x\}$ and $W_{Y|X,S}(y\,|\,x,s_1) = \mathbf{1}\{y \neq x\}$. The no-adversary channel is the noiseless binary channel, with $C_{s_0} = 1$ bit/use.

I-overwritable. For any $(x,x') \in \mathscr{X}^2$, the oblivious mixing $\mathsf{Q}(s_0) = \mathbf{1}\{x = x'\}$, $\mathsf{Q}(s_1) = \mathbf{1}\{x \neq x'\}$ matches $W_{Y|X,S}(\cdot \mid x', s_0)$, since under input x the channel delivers x' with probability one when the state is chosen this way.

Failure of distribution-overwritability. For target $x' = 0$ and input distribution $\mathsf{P}_X(0) = p \in (0,1)$, an oblivious state distribution $\mathsf{Q}(s_0) = q$ induces

$$\Pr(Y=0) \;=\; p\,q \;+\; (1-p)(1-q),$$

which equals 1 only at $\{p,q\} \in \{(0,1),(1,0)\}$. Both points exclude $p \in (0,1)$, so no choice of Q matches the no-adversary marginal at $x' = 0$ when the input is genuinely randomized.

Capacity consequence. By Theorem 7, $C^{\bullet}_{\text{auth,det}} = 0$, while Theorems 5 and 6 give $C^{\bullet}_{\text{auth,stoch}} = C_{s_0} = 1$ bit/use.

Appendix G
$\mathrm{AZBSC}(\delta, \alpha)$: distribution-overwritable but not obliviously overwritable

The AZBSC channel and the separation it exhibits are from [5]; we reprise the verification here for completeness.

The $\mathrm{AZBSC}(\delta, \alpha)$ has $\mathscr{X} = \mathscr{Y} = \{0, 1\}$, $\mathscr{S} = \{s_0, s_{1,0}, s_{1,1}\}$, and parameters $0 < \alpha < \delta < 1/2$. Under state s_0, $W_{Y|X,S}(\cdot\,|\,\cdot, s_0)$ is a $\mathrm{BSC}(\delta)$; under $s_{1,x}$, $W_{Y|X,S}(y\,|\,x, s_{1,x}) = \mathbf{1}\{y = x\}$; and

$$W_{Y|X,S}(y\,|\,x, s_{1,x\oplus 1}) = \alpha\,\mathbf{1}\{y = x\} + (1-\alpha)\mathbf{1}\{y \neq x\}\,.$$

Distribution-overwritability. For $\mathsf{P}_X(0) = p$ and target $x' = 0$, the strategy

$$\mathsf{Q}(s_0) = 0, \quad \mathsf{Q}(s_{1,0}) = \tfrac{1-\delta-p\alpha}{1-\alpha}, \quad \mathsf{Q}(s_{1,1}) = \tfrac{\delta-(1-p)\alpha}{1-\alpha}$$

satisfies $\mathbb{E}_{X,S} W_{Y|X,S}(\cdot\,|\,X, S) = W_{Y|X,S}(\cdot\,|\,0, s_0)$.

Failure of oblivious overwritability. An oblivious $\mathsf{Q}(s_{1,0}) = q_0$, $\mathsf{Q}(s_{1,1}) = q_1$ matching $W_{Y|X,S}(\cdot\,|\,0, s_0)$ for both $x = 0$ and $x = 1$ forces $q_0 + q_1 = (1-2\delta)/(1-2\delta-\alpha) \notin [0, 1]$ for any admissible (δ, α).

Capacity consequence. By Theorems 5 and 6, $C^{\bullet}_{\mathrm{auth,stoch}} = C_{s_0}$, the no-adversary capacity of the $\mathrm{BSC}(\delta)$, while the channel is not obliviously overwritable so the dichotomy of [2] would not yield this conclusion.